

AgriSafe: Intelligent Farming Agent Under Noisy and Misleading Sensor Conditions

Aseel Aboramdan
Ola Abuara
Lamar Alzahrani
Computer Science Department

Abstract

AgriSafe is an intelligent agent designed to operate in a simulated agricultural environment under noisy and misleading sensor conditions. The system models a smart farming scenario where an AI agent monitors environmental factors such as soil moisture, temperature, and plant health to make automated decisions. To reflect real-world challenges, the environment introduces adversarial noise into sensor readings, simulating faulty or unreliable data. The agent is required to maintain plant health while avoiding incorrect actions caused by misleading inputs. A rule-based reasoning approach is implemented to ensure robustness under uncertainty. The agent evaluates multiple environmental conditions before selecting actions such as watering, waiting, or triggering alerts. The system is tested in both normal and adversarial environments to evaluate performance differences. Results show that although noisy conditions slightly reduce performance, the agent remains stable and avoids critical failures such as plant death or excessive resource usage. This demonstrates that even simple rule-based intelligent agents can perform effectively under uncertainty when designed with robustness in mind. The project highlights the importance of reasoning under noisy data in artificial intelligence and its applications in smart agriculture systems.

1 Introduction

Smart agriculture systems depend on sensor data to automate decision-making. However, real-world sensors are often noisy or unreliable. This creates challenges for intelligent agents that rely on accurate observations.

AgriSafe simulates this problem by introducing misleading sensor data and evaluating how an AI agent responds under uncertainty.

2 Related Work

Intelligent agents in agriculture are widely used in precision farming systems. Many approaches rely on rule-based logic or machine learning. However, most assume accurate sensor data.

This project focuses on robustness under uncertainty using a rule-based agent.

3 System Design

3.1 Environment Model

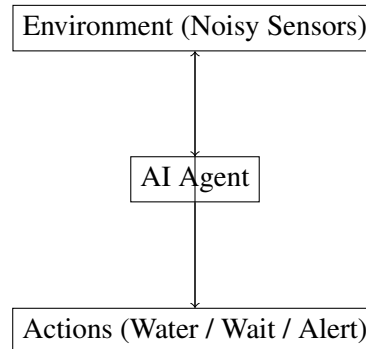
The environment simulates:

- Soil moisture (0–100%)

- Temperature (°C)
- Plant health (0–100)

Noise is introduced by distorting soil moisture readings.

3.2 Agent Architecture



4 Agent Decision Algorithm

```

If soil moisture is low -> WATER
If moisture is borderline and temperature is low -> WAIT
If temperature is very high -> WATER
If plant health is low -> ALERT
Else -> WAIT
  
```

5 Methodology

The system is tested in two environments:

- Normal environment (accurate data)
- Adversarial environment (noisy data)

Performance is measured using average plant health over time.

6 Implementation Details

The system was implemented in Python using a modular structure. The project includes a simulation environment, an intelligent agent, and an evaluation module. The environment introduces adversarial noise to simulate real-world sensor uncertainty. The agent uses rule-based reasoning to make decisions under uncertainty.

The repository includes a README file with installation instructions and a requirements file listing dependencies.

7 Results and Discussion

In the normal environment, the agent maintains stable plant health. In the noisy environment, performance decreases slightly due to misleading sensor data.

However, the agent avoids major failures, showing robustness under uncertainty.

8 Individual Contributions

Aseel Aboramdan:

- Designed the AI agent logic and decision rules
- Developed the simulation environment structure
- Wrote the introduction and system design sections

Ola Abuara:

- Implemented sensor noise and adversarial conditions
- Assisted in testing normal vs noisy environments
- Contributed to results and discussion section

Lamar Alzahrani:

- Worked on evaluation metrics and performance analysis
- Helped refine agent decision rules
- Contributed to conclusion and future work sections

9 Conclusion

AgriSafe demonstrates a robust intelligent agent capable of operating under noisy sensor conditions. The system shows that rule-based AI can still perform effectively under uncertainty.

10 Future Work

- Reinforcement learning integration
- Probabilistic reasoning methods
- Multi-agent farming simulation

11 References

- Russell, S., Norvig, P. Artificial Intelligence: A Modern Approach.
- OpenAI Gym Documentation.